

MSTAR PRACTICE WORKSHEET**Lesson 2-6: Divide Multi-Digit Numbers Using an Algorithm**

Grade 6 Mathematics · Standard 6.NOS.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**What is $936 \div 12$?

- ☐ A 68
- ☐ B 76
- ☒ C 78
- ☐ D 87

$12 \times 78 = 936$. You can verify: $12 \times 70 = 840$, $12 \times 8 = 96$, $840 + 96 = 936$.

Question 2 **SELECTED RESPONSE — CORRECT: C**What is $2,485 \div 5$?

- ☐ A 487
- ☐ B 496
- ☒ C 497
- ☐ D 507

$5 \times 497 = 2,485$. Running the cycle: $24 \div 5 \rightarrow 4$ with 4 left, $48 \div 5 \rightarrow 9$ with 3 left, $35 \div 5 \rightarrow 7$ with 0 left, so the quotient is 497 and the remainder is 0.

Question 3 SELECTED RESPONSE — CORRECT: B

What is $756 \div 9$?

- ☐ A 83
- ☒ B 84
- ☐ C 86
- ☐ D 94

$9 \times 84 = 756$. Running the cycle: $75 \div 9 \rightarrow 8$ with 3 left, then $36 \div 9 \rightarrow 4$ with 0 left, so the quotient is 84 and the remainder is 0.

Question 4 SELECTED RESPONSE — CORRECT: C

What is $1,125 \div 9$?

- ☐ A 115
- ☐ B 124
- ☒ C 125
- ☐ D 135

$9 \times 125 = 1,125$. Running the cycle: $11 \div 9 \rightarrow 1$ with 2 left, $22 \div 9 \rightarrow 2$ with 4 left, $45 \div 9 \rightarrow 5$ with 0 left, so the quotient is 125.

Question 5 SELECTED RESPONSE — CORRECT: C

A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

- ☐ A 146 boxes
- ☐ B 155 boxes
- ☒ C 156 boxes
- ☐ D 166 boxes

$2,184 \div 14 = 156$. Check: $14 \times 156 = 14 \times 150 + 14 \times 6 = 2,100 + 84 = 2,184$.

Question 6

SELECTED RESPONSE — CORRECT: C

What is $4,896 \div 12$?

- ☐ A 48
- ☐ B 380
- ☒ C 408
- ☐ D 480

$$12 \times 408 = 4,896.$$

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

Use the long division algorithm to find $936 \div 4$.

- ☐ A 209
- ☐ B 232
- ☒ C 234
- ☐ D 243

$9 \div 4 = 2$ remainder 1; bring down 3 to make 13. $13 \div 4 = 3$ remainder 1; bring down 6 to make 16. $16 \div 4 = 4$. The quotient is 234.

- **A:** A digit was skipped. After 9 divided by 4 leaves remainder 1, bring the 3 down to make 13, which gives the tens digit.
- **B:** The last step is 16 divided by 4, which is 4. Check the final digit after bringing down the 6.
- **D:** The tens and ones digits are swapped. Work left to right: 2, then 3, then 4.

Part B — correct answer: B

In the D-M-S-B cycle, what does the SUBTRACT step tell you?

- ☐ A How many times the divisor fits into the whole dividend.
- ☒ B How much of the current value is left over before the next digit is brought down.
- ☐ C Whether the quotient will be even.
- ☐ D The final remainder of the entire problem.

Subtracting the product from the current partial dividend leaves the remainder at that place, which the next digit is appended to.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Dropped Zero

Andre computed $618 \div 3$. He wrote: ' $6 \div 3 = 2$, then $1 \div 3$ does not go so I skip it, then $8 \div 3 = 2$ remainder 2. My answer is 22 remainder 2.'

Explain Andre's misconception and give the correct quotient.

Model answer: $618 \div 3 = 206$. When $1 \div 3$ does not go, the quotient gets a 0 in the tens place and the 1 stays to make 18 with the next digit; $18 \div 3 = 6$. Skipping the place entirely loses the value of that digit.

2 points	Student explains that when a digit is too small the quotient gets a 0 in that place and the digit is carried to the next step (1 becomes 18), and gives 206.
1 point	Student gives 206 without explaining the placeholder zero.
0 points	Student agrees with Andre or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

The station must load 3,360 ration packs equally into 14 storage pods. Solve it with long division. Name each step — divide, multiply, subtract, bring down — and check your answer by multiplying.

Model answer: Long division on $3,360 \div 14$ runs the cycle three times: $33 \div 14 \rightarrow 2$ ($2 \times 14 = 28$, leaving 5), bring down the 6 $\rightarrow 56 \div 14 \rightarrow 4$ ($4 \times 14 = 56$, leaving 0), bring down the 0 $\rightarrow 0 \div 14 \rightarrow 0$, so a 0 must be written in the ones place. The quotient is 240 ration packs per pod, and $14 \times 240 = 3,360$ proves none were left over.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.